Weekly Progress Log – Week 5

Project Title: Data Science with Python

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Week: 5

Objectives for Week

- Learn to visualize distributions, relationships, and trends using Matplotlib and Seaborn.
- Perform Exploratory Data Analysis (EDA) on the Titanic dataset cleaned during Week 3.
- Use histograms, box plots, scatter plots, and heatmaps to explore different features.
- Compare passenger survival patterns using descriptive statistics and group summaries.
- Identify three to five meaningful findings supported by charts and explain them in simple language.
- Prepare a documented Jupyter Notebook as the Week 5 EDA deliverable.

Work Completed

- Loaded the Week 3 cleaned Titanic passenger dataset into Pandas for analysis (891 rows and 11 columns).
- Used df.head(), df.shape, df.columns, df.info(), and df.describe() to review the dataset structure and descriptive statistics.
- Checked missing values and duplicate records; the cleaned file contained no missing values or exact duplicate rows.
- Created an age-distribution histogram with a Kernel Density Estimation (KDE) curve to visualize the distribution of passenger ages.
- Built a box plot of ticket fares by passenger class to compare distributions and identify potential outliers.
- Created an age-versus-fare scatter plot, colored by survival status, to explore passenger-level relationships.
- Calculated correlations among numerical features and displayed them using an annotated heatmap.
- Calculated the overall survival rate (38.38%) and compared survival rates across gender and passenger class.
- Prepared bar charts for survival rate by gender, class, and family-size category.
- Created family-size groups (alone, 2–4 people, and 5 or more people) for a clearer survival comparison.
- Examined ticket fare statistics, including the median fare of approximately 14.45 and maximum of approximately 512.33.
- Documented five written insights, explained each chart in plain language, and included a conclusion with analysis limitations.
- Organized the work in separate code and Markdown cells in the EDA_Project.ipynb notebook.

Challenges Faced

- Selecting appropriate chart types to represent distributions, comparisons, and relationships.

- Interpreting the unusual fare values shown as potential outliers on a box plot.
- Interpreting positive and negative correlations and understanding why correlation does not necessarily imply causation.
- Recognizing that filling missing ages with the median in Week 3 affected the age distribution.
- Summarizing statistical results as clear, non-technical insights supported by the visualizations.

Solutions Implemented

- Used histograms for distributions, box plots for fare spread, scatter plots for variable relationships, and heatmaps for correlations.
- Used Pandas descriptive statistics and class comparisons to interpret unusually high ticket fares.
- Calculated group survival rates with groupby() and mean(), then used bar charts to explain differences.
- Used Markdown explanations to describe potential outliers, explain how replacing missing ages with the median affected the age distribution, and clarify that correlation does not prove causation.
- Grouped family size into understandable categories and explained findings using percentages and simple sentences.
- Documented the analysis, five insights, and conclusion directly inside the Jupyter Notebook.

Individual Tasks

Responsibility	Description of Contribution	Remark
Data Loading	Loaded the cleaned Titanic CSV in Pandas and reviewed the available fields.	Prepared the Week 3 data for EDA.
Data Validation	Checked dimensions, data types, missing values, duplicates, and basic statistics.	Confirmed 891 rows, 11 columns, and no missing values.
Age Distribution	Created a histogram with a KDE curve to visualize passenger ages.	Examined the age distribution and identified the effect of replacing missing ages with the median.
Fare Comparison	Created a box plot of ticket fare by passenger class.	Observed fare variation and potential outliers.
Scatter Plot	Plotted age against fare with points identified by survival status.	Explored relationships between numerical values.
Correlation Analysis	Computed numerical correlations and displayed an annotated heatmap.	Learned to interpret positive and negative correlations.
Survival Analysis	Calculated overall survival and grouped results by gender and passenger class; created bar charts.	Explained differences in survival percentages.
Family Size Analysis	Created family-size groups and compared their survival rates in a bar chart.	Identified differences among alone, small-family, and large-family groups.
Insights and Reporting	Wrote five insights, chart explanations, limitations, and a conclusion in the notebook.	Prepared the Week 5 EDA deliverable for review.